

5 Complete the following paragraphs using the most appropriate word or words.

The theory of evolution describes a process that can lead to the formation of new species from pre-existing species over

DNA sequence data of different species can be compared to show evolutionary relationships. Two species that have a more recent common ancestor share more in the DNA nucleotide sequences of their genomes than two species that are more distantly related.

Mitochondrial DNA can also be used in the study of evolutionary relationships. Mitochondrial DNA is inherited only from the female gamete, and its nucleotide sequence is unaffected by during the production of gametes.

DNA sequence data can be stored in large biological, allowing faster comparison of the nucleotide sequences of genomes using computer software. DNA sequence data can also be used to predict the sequences of proteins produced by a species.

A can be used to detect many different mRNA molecules at the same time in studies that compare gene expression between different species.